


# *Standard Operating Procedure HAAS* CNC Mill

## HAAS VF2-SS CNC Mill

The **Haas VF2-ss** is a high-speed, 3-axis CNC vertical machining center featuring a **12,000-RPM** inline direct-drive spindle and high-speed rapids of **1400 IPM** (inch per minute). This machine is an automated, high-precision tool. Unlike manual equipment, it can initiate rapid motion at any time during a program cycle.

## Potential Hazards

- Be aware of the following potential hazards when working with band saws:
    - Mechanical: Rapid axis movement (1400 ipm) and the Side-Mount Tool Changer (SMTC) present severe pinch and crush points. Never place hands between the tool changer and the spindle head.
    - Ejection: Improperly clamped parts or oversized stock can be ejected with deadly force, potentially puncturing the safety enclosure.
    - Electrical: High voltage 220V is present in the control cabinet. The cabinet must remain locked and accessible only to qualified personnel.
    - Chemical: CNC coolants can be corrosive to eyes and skin; avoid ingesting or inhaling mist.

## Before Use

- Warm**-Up Cycle**: If the machine has been idle for >24 hours, you **must** run the Spindle Warm-Up program (typically 20 minutes) to ensure proper bearing lubrication.
- **Balanced Tooling**: For operations at 10,000 RPM or higher, you must use balanced tool holders (Grade **G2.5** or better per ANSI S2.19). **WE DO NOT OWN ANY**
- **Clearance Check**: Ensure the work zone is clear of loose tools and the Z-axis height is verified for all tool lengths.
- **Coolant System Preparation**: Before starting the machine, inspect and clean the coolant tank intake filter and the chip tray screen to ensure unobstructed flow back to the reservoir.




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## During Use

- **First-Run Safety**: Maintain a hand on the **Emergency Stop** button during the first run of any new program. Use 5% rapid overrides until tool paths are verified. Low Initial feed rate is also recommended to verify correct Speed and feed rate.
- **Enclosure**: Keep doors closed and guards in place during all automatic operations.
- **Work holding**: Ensure pressure is set correctly and the workpiece is securely clamped. Do not machine parts larger than the work holding capacity.

## After Use

- **Chip Clearing**: Use washdown hoses or brushes only when all machine operation has stopped & Emergency Stop is depressed.
- **Coolant Tray Filtration**: After usage and chip washdown, the primary coolant tray filter **must** be cleared of all fine debris and sludge to prevent pump cavitation and sensor failure. Shake out the Cloth net filter.
- **Internal Cleanup**: Do not use compressed air for cleaning; it forces fine particulates into precision bearings and under way-covers.
- **Power Down**: Follow the pendant "Power Off" sequence before engaging the main circuit breaker.

## Emergency Equipment and Procedures

- **E-Stop**: The large red circular button on the pendant immediately stops all motion. In a power loss event, be aware that the spindle will take significantly longer to coast to a stop.
- **Incident Reporting**: Immediately report any tool crashes, enclosure damage, or near misses to the Laboratory Project Supervisor via phone.
- Know the location of the nearest emergency equipment and items, such as the emergency stop/shut-off for the specific machine, eye wash, first aid kit, fire extinguishers, fire alarm pull stations, and emergency exits.



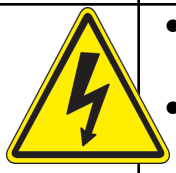

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## Appendix B - Job Hazard Analysis

| Work Duties | Potential Hazards | Controls |
|---|---|---|
| Operating the Machine | **Flying Debris**  | • Safety glasses are required while o<br>• Machine mounted guards must b an appropriate manner |
| Operating the Machine | **Pinch, Crush and Entanglement Hazards**  | • Use caution while handling heavy devices |




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|---|---|---|
| Operating the Machine | **Entanglement, Lacerations and Flying Debris Hazards** | • During Tool Changes, machine mu<br>• Operator must be trained in th use of the brake to prevent startup<br>• Never reach towards the operation while machine is runn |
| Operating the Machine | **Fire**  | • Operators are trained to shut d machine in accordance with outlined in the machine-specific SOP<br>• All combustibles must be clear of th metal shavings, combustible dust debris must be cleaned up appropria<br>• Class D extinguishers must be acces times |
| Operating the Machine | **Slips, Trips and Falls**  | • Proper housekeeping must be m at all times, including cleaning immediately |
| Machine Service | **Electrocution/Accidental Start up**  | • Always follow Lock out / Tag out control of hazardous energy<br>• Equipment needing serviced shall out immediately by authorized per<br>• Service should be completed in the manufacturer requirements b and authorized personnel or a cont |
| Post-Operation Housekeeping | **Lacerations or Foreign Body in Eyes**  | • When cleaning up safety glasses a must be worn at all times<br>• NEVER attempt cleaning up swarf/cl machine is running<br>• Gloves should be used with cautio machinery, as they may pre entanglement hazard |